# Solutions

## Maths

Q1. Let $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x)=\begin{cases}x+2, & x<1\\ 4, & x=1\\ x^2+1, & x>1\end{cases}$ and let $g(x)=1-x^2$. Then $\lim\limits_{x\to 0} f(g(x))$ is equal to:

(A) 4 (B) 3 (C) 2 (D) Does not exist

Ans. B

Sol. RHL = f(g($0^+$)=f($1^-$) = 3

LHL = f(g($0^-$)=f($1^-$) = 3

$\lim\limits_{x\to 0} f(g(x))$ = 3

Q2. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=(a^2-3a+2)\cos x+\ln\left(x+\sqrt{x^2+1}\right)+(b-1)x^3$ where $a,b\in\mathbb{R}$. If $f(x)$ is an odd function which is also one-one (injective) on $\mathbb{R}$, then the set of all possible ordered pairs $(a,b)$ is :

(A) $\{1,2\}\times[1,\infty)$ (B) $\{1,2\}\times(1,\infty)$ (C) $[1,2]\times[1,\infty)$ (D) $\{1\}\times[1,\infty)$

Ans. A

Sol. For $f(x)$ to be an odd function, $f(-x)=-f(x)$ must hold for all $x\in\mathbb{R}$.

Note that $\ln\left(-x+\sqrt{x^2+1}\right)=\ln\left(\dfrac{1}{x+\sqrt{x^2+1}}\right)=-\ln\left(x+\sqrt{x^2+1}\right)$.

Thus, $f(-x)=(a^2-3a+2)\cos x-\ln\left(x+\sqrt{x^2+1}\right)-(b-1)x^3$

Setting $f(-x)=-f(x)$ yields: $2(a^2-3a+2)\cos x=0\quad\forall x\in\mathbb{R}$

$\implies a^2-3a+2=0\implies a=1$ or $a=2$

For $a\in\{1,2\}$, the function reduces to: $f(x)=\ln\left(x+\sqrt{x^2+1}\right)+(b-1)x^3$

Differentiating with respect to $x$: $f'(x)=\dfrac{1}{\sqrt{x^2+1}}+3(b-1)x^2$

Since $f'(0)=1>0$, for $f(x)$ to be one-one (strictly monotonic) on $\mathbb{R}$, we must have $f'(x)\geq 0$ for all $x\in\mathbb{R}$.

- If $b\geq 1$, then $3(b-1)x^2\geq 0$, giving $f'(x)>0$ for all $x\in\mathbb{R}$. Hence $f(x)$ is strictly increasing and one-one.

- If $b < 1$, then as $x \to \infty$, $f'(x) \to -\infty$, which causes $f'(x)$ to change sign and renders $f(x)$ many-one.
  Therefore, $b \in [1, \infty)$.

  The set of all ordered pairs $(a, b)$ is $\{1, 2\} \times [1, \infty)$.

Q3. Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function defined by
$f(x) = \left|\sin\left(\frac{2\pi x}{3}\right)\right| + \left|\cos\left(\frac{\pi x}{2}\right)\right|$
Let $T$ be the fundamental period of $f(x)$. Consider the sets $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, \ldots, T\}$. The total number of functions $g : A \to B$ such that $g(1) < g(2)$, $g(3) \leq g(4)$, and $g$ is a many-one function is :

(A) 135 (B) 180 (C) 225 (D) 315

Ans. C

Sol. The fundamental period T of f(x) = 6

Thus, set $B = \{1, 2, 3, 4, 5, 6\}$.

Next, we count the functions $g : A \to B$ where $A = \{1, 2, 3, 4\}$.

Number of ways to choose $(g(1), g(2))$ such that $g(1) < g(2)$:

We select 2 distinct elements from $B$ in $\binom{6}{2} = 15$ ways.

Number of ways to choose $(g(3), g(4))$ such that $g(3) \leq g(4)$:

This is equivalent to choosing 2 elements from $B$ with repetition allowed, which is given by $\binom{6+2-1}{2} = \binom{7}{2} = 21$ ways.

Total number of functions satisfying $g(1) < g(2)$ and $g(3) \leq g(4)$ without restriction on injectivity: $N_{\text{total}} = 15 \times 21 = 315$

The number of injective functions is: $N_{\text{injective}} = \binom{6}{4} \times \binom{4}{2} = 15 \times 6 = 90$

Therefore, the number of many-one functions is: $N_{\text{many-one}} = N_{\text{total}} - N_{\text{injective}} = 315 - 90 = 225$

Q4. Let $f : \mathbb{R} \setminus \{2\} \to \mathbb{R}$ be a function defined as $f(x) = \begin{cases} [x] + \text{sgn}(\sin(\pi x)) + \frac{\tan(\pi x)}{x-2}, & x < 2 \\ \frac{2^x - 4}{x-2} & x > 2 \end{cases}$
where $[\cdot]$ denotes the greatest integer function and $\text{sgn}(\cdot)$ denotes the signum function.

Let $g(x) = \dfrac{x \sin x}{1 - \cos x}$ for $x \neq 0$. Then $\lim\limits_{x \to 0} f(g(x))$ is equal to :

(A) $1 + \pi$ (B) $2 + \pi$ (C) $4 \ln 2$ (D) $\pi$

Ans. D

Sol. First, analyze the behavior of $g(x)$ as $x \to 0$:

$$g(x) = \frac{x \sin x}{1 - \cos x} = \frac{x\left(2\sin\frac{x}{2}\cos\frac{x}{2}\right)}{2\sin^2\frac{x}{2}} = 2 \cdot \frac{\frac{x}{2}}{\tan\frac{x}{2}}$$

Therefore, as $x \to 0$, $g(x) \to 2^-$.

To evaluate $\lim\limits_{x\to 0} f(g(x))$, we substitute $t = g(x)$,

$$\lim_{t\to 2^-}\left([t] + \operatorname{sgn}(\sin(\pi x)) + \frac{\tan(\pi t)}{(t-2)}\right)$$

$$\lim_{t\to 2^-}\left([t] + \operatorname{sgn}(\sin(\pi x)) + \frac{\tan(\pi(t-2))}{\pi(t-2)} \cdot \pi\right) = 1 - 1 + \pi = \pi$$

Q5. Consider the function $f : [-1, 1] \to \left[\frac{1}{3}, 3\right]$ defined by

$$f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$$

Which of the following statement(s) is/are **CORRECT**?

(A) The equation $f(x) = f^{-1}(x)$ has exactly three real solutions in the common domain of $f$ and $f^{-1}$.

(B) $f(x)$ is a bijective function, and its inverse function $f^{-1} : \left[\frac{1}{3}, 3\right] \to [-1, 1]$ satisfies $f^{-1}(1) = 0$.

(C) The equation $f(x) = f^{-1}(x)$ has a unique real solution $x = \alpha$ in the common domain of $f$ and $f^{-1}$, where $\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right)$.

(D) The range of the function $g(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$ for $x \in \mathbb{R}$ is $\left[\frac{1}{3}, 3\right]$.

Ans. B, C, D

Sol. **Range of $g(x)$ on $\mathbb{R}$:**

Let $y = \frac{x^2 - x + 1}{x^2 + x + 1}$. Rearranging gives: $(y-1)x^2 + (y+1)x + (y-1) = 0$

For $y = 1$, $x = 0 \in \mathbb{R}$.

For $y \neq 1$, since $x \in \mathbb{R}$, the discriminant $D \geq 0$:

$$D = (y+1)^2 - 4(y-1)^2 \geq 0 \implies (3-y)(3y-1) \geq 0 \implies y \in \left[\frac{1}{3}, 3\right]$$

$$f'(x) = \frac{2(x^2-1)}{(x^2+x+1)^2}$$

For $x \in (-1, 1)$, $f'(x) < 0$, so $f(x)$ is strictly decreasing on $[-1, 1]$.

$f(-1) = 3$ and $f(1) = \frac{1}{3}$.

Since $f$ is continuous and strictly decreasing, its image is $\left[\frac{1}{3}, 3\right]$, making $f$ bijective.

Also, $f(0) = 1 \implies f^{-1}(1) = 0$.

Thus, statement (B) is correct.

**Solving $f(x) = f^{-1}(x)$:**

The common domain of $f$ and $f^{-1}$ is $[-1,1] \cap \left[\frac{1}{3}, 3\right] = \left[\frac{1}{3}, 1\right]$.

Setting $f(x) = x$: $\frac{x^2 - x + 1}{x^2 + x + 1} = x \implies x^3 + 2x - 1 = 0$

Let $h(x) = x^3 + 2x - 1$.

$h\left(\frac{1}{3}\right) = \frac{1}{27} + \frac{2}{3} - 1 = -\frac{8}{27} < 0$; $h\left(\frac{1}{2}\right) = \frac{1}{8} + 1 - 1 = \frac{1}{8} > 0$

By Intermediate Value Theorem, the unique root $\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right)$.

Thus, statement (C) is correct

Q6. Let $f : [-2,2] \to \mathbb{R}$ be a function defined by $f(x) = \begin{cases} \max\left(x+2, x^2\right) + ax, & -2 \le x < 0 \\ \min\left(2-x, 2-x^2\right) + bx, & 0 \le x \le 2 \end{cases}$ where $a, b \in \mathbb{R}$. Which of the following statement(s) is/are **CORRECT** ?

(A) $f(x)$ is continuous on $[-2,2]$ for all real values of $a$ and $b$.

(B) The function $f(x)$ is non-differentiable at exactly two points in $(-2,2)$ if and only if $b - a = 2$.

(C) If $a = 3$ and $b = -2$, then by the Intermediate Value Theorem, the equation $f(x) = 0$ has at least one real root in $(-2,0)$ and at least one real root in $(0,2)$.

(D) There exists at least one pair of real numbers $(a,b)$ such that $f(x)$ is differentiable at all points in $(-2,2)$.

Ans. A, B, C

Sol. $f(x) = \begin{cases} x^2 + ax, & -2 \le x \le -1 \\ x + 2 + ax, & -1 < x < 0 \\ 2 - x + bx, & 0 \le x \le 1 \\ 2 - x^2 + bx, & 1 < x \le 2 \end{cases}$

**Check for Continuity**

At $x = -1$: $\lim\limits_{x \to -1^-} f(x) = 1 - a = \lim\limits_{x \to -1^+} f(x) = f(-1)$.

At $x = 0$: $\lim\limits_{x \to 0^-} f(x) = 2 = \lim\limits_{x \to 0^+} f(x) = f(0)$.

At $x = 1$: $\lim\limits_{x \to 1^-} f(x) = 1 + b = \lim\limits_{x \to 1^+} f(x) = f(1)$.

Since each piece is polynomial and continuous, $f(x)$ is continuous on $[-2,2]$ for all $a, b \in \mathbb{R}$.

Hence, **(A) is correct**.

Evaluating the left-hand and right-hand derivatives at critical transition points $x = -1, 0, 1$:

At $x = -1$: $LHD = -2 + a$, $RHD = 1 + a$.

Since $LHD \neq RHD$, $f(x)$ is **never differentiable** at $x = -1$.

At $x = 1$: $LHD = -1 + b$, $RHD = -2 + b$.

Since $LHD \neq RHD$, $f(x)$ is **never differentiable** at $x = 1$.

At $x = 0$: $LHD = 1 + a$, $RHD = -1 + b$.

Differentiable at $x = 0 \iff 1 + a = -1 + b \iff b - a = 2$.

Therefore:

If $b-a=2$, non-differentiability points in $(-2,2)$ are exactly $\{-1,1\}$ (2 points).

If $b-a\neq 2$, non-differentiability points in $(-2,2)$ are $\{-1,0,1\}$ (3 points).

Hence, **(B) is correct**, and **(D) is incorrect** (since $x=-1,1$ are always non-differentiable points for any pair $(a,b)$).

For $a=3$ and $b=-2$: - $f(-2)=(-2)^2+3(-2)=-2<0$. $f(0)=2>0$.

Since $f(x)$ is continuous on $[-2,0]$ and $f(-2)\cdot f(0)<0$, by IVT there exists at least one $c_1\in(-2,0)$ such that $f(c_1)=0$.

$f(2)=2-2^2+(-2)(2)=-6<0$.

Since $f(x)$ is continuous on $[0,2]$ and $f(0)\cdot f(2)<0$, by IVT there exists at least one $c_2\in(0,2)$ such that $f(c_2)=0$.

Hence, **(C) is correct**.

Q7. Let $f:[0,3]\to\mathbb{R}$ be a function defined by

$$f(x)=\begin{cases}\max_{0\le t\le x}(t^3-3t+2), & \text{if } 0\le x\le 2\\ x^2+5x-10, & \text{if } 2<x\le 3\end{cases}$$

Which of the following statements is/are **CORRECT** ?

(A) $f(x)$ is non-differentiable at exactly one point in $(0,3)$.

(B) The equation $f(x)=2x$ has exactly two solutions in the interval $[0,3]$.

(C) $f(x)$ is discontinuous at $x=\sqrt{3}$.

(D) By the Intermediate Value Theorem, there exists at least one $c\in(1,2)$ such that $f(c)=5$.

Ans. A, B

Sol. $f(x)=\begin{cases}2, & \text{if } 0\le x\le\sqrt{3}\\ x^3-3x+2, & \text{if } \sqrt{3}<x\le 2\\ x^2+5x-10, & \text{if } 2<x\le 3\end{cases}$

Continuity Check: At $x=\sqrt{3}$: $\lim\limits_{x\to\sqrt{3}^-}f(x)=2$ and $\lim\limits_{x\to\sqrt{3}^+}f(x)=(\sqrt{3})^3-3\sqrt{3}+2=2=f(\sqrt{3})$

.

Continuous. (Option C is incorrect).

At $x=2$: $\lim\limits_{x\to 2^-}f(x)=2^3-3(2)+2=4$ and $\lim\limits_{x\to 2^+}f(x)=2^2+5(2)-10=4=f(2)$.

Continuous.

Thus, $f(x)$ is continuous everywhere on $[0,3]$.

Differentiability Check: At $x=\sqrt{3}$: $f'_-(\sqrt{3})=0$, whereas $f'_+(\sqrt{3})=\dfrac{d}{dx}(x^3-3x+2)\Big|_{x=\sqrt{3}}=3(\sqrt{3})^2-3=6$.

Since $f'_-(\sqrt{3})\neq f'_+(\sqrt{3})$, $f(x)$ is non-differentiable at $x=\sqrt{3}$.

At $x=2$: $f'_-(2)=3(2)^2-3=9$ and $f'_+(2)=2(2)+5=9$.

Since $f'_-(2)=f'_+(2)$, $f(x)$ is differentiable at $x=2$.

Therefore, $f(x)$ is non-differentiable at exactly one point ($x=\sqrt{3}$) in $(0,3)$. (Option A is correct).

Solving $f(x) = 2x$ on $[0, 3]$:

On $[0, \sqrt{3}]$: $2 = 2x \implies x = 1 \in [0, \sqrt{3}]$.

On $(\sqrt{3}, 2]$: $x^3 - 3x + 2 = 2x \implies x^3 - 5x + 2 = 0$.
At $x = 2$, $2^3 - 5(2) + 2 = 0 \implies x = 2 \in (\sqrt{3}, 2]$.
For $x \in [\sqrt{3}, 2]$, $\dfrac{d}{dx}(x^3 - 5x + 2) = 3x^2 - 5 \geq 4 > 0$, so no other roots exist in this subinterval.

On $(2, 3]$: $x^2 + 5x - 10 = 2x \implies x^2 + 3x - 10 = 0 \implies (x + 5)(x - 2) = 0$, no root in $(2, 3]$.
Thus, the equation $f(x) = 2x$ has exactly two solutions ($x = 1$ and $x = 2$) in $[0, 3]$. (Option B is correct).
Intermediate Value Theorem : On $[1, 2]$, $f(x)$ is continuous and ranges from $f(1) = 2$ to $f(2) = 4$.
Since $5 \notin [2, 4]$, there exists no $c \in (1, 2)$ such that $f(c) = 5$. (Option D is incorrect).

Q8. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} \frac{a\cos x + bx\sin x + ce^x - 1 - x}{x^3}, & x < 0 \\ d, & x = 0 \\ \frac{e^{kx} - 1 - kx - mx^2}{x^3}, & x > 0 \end{cases}$$

where $a, b, c, d, k, m \in \mathbb{R}$.

Which of the following statement(s) is/are **CORRECT** ?

(A) If $a = 0, b = -\dfrac{1}{2}, c = 1, k = 1, m = \dfrac{1}{2}$, and $d = \dfrac{1}{6}$, then $f(x)$ is continuous at $x = 0$.

(B) If $a = 0, b = -\dfrac{1}{2}, c = 1, k = 1, m = \dfrac{1}{2}$, and $d \neq \dfrac{1}{6}$, then $f(x)$ has a removable isolated point discontinuity at $x = 0$.

(C) If $a = 0, b = -\dfrac{1}{2}, c = 1, k = 2$, and $m = 2$, then $f(x)$ has a non-removable jump discontinuity at $x = 0$ of magnitude $\dfrac{1}{6}$ for any real value of $d$.

(D) If $a = 0, c = 1$, and $b = \dfrac{1}{2}$, then $f(x)$ has an essential infinite discontinuity at $x = 0$.

Ans. A, B, D

Sol. **Left-Hand Limit (LHL):** For $x < 0$, Using expansion numerator becomes

$$(a + c - 1) + x(c - 1) + x^2\left(b - \frac{a}{2} + \frac{c}{2}\right) + x^3\left(\frac{c}{6}\right) + O(x^4)$$

For $\lim\limits_{x \to 0^-} f(x)$ to exist finitely:

1. $a + c - 1 = 0$
2. $c - 1 = 0 \implies c = 1 \implies a = 0$
3. $b - \dfrac{a}{2} + \dfrac{c}{2} = 0 \implies b + \dfrac{1}{2} = 0 \implies b = -\dfrac{1}{2}$

When $a = 0, b = -\dfrac{1}{2}, c = 1$: LHL $= \lim\limits_{x \to 0^-} f(x) = \dfrac{c}{6} = \dfrac{1}{6}$

**Right-Hand Limit (RHL):** For $x > 0$, Using expansion numerator becomes

$$e^{kx} - 1 - kx - mx^2 = x^2\left(\frac{k^2}{2} - m\right) + x^3\left(\frac{k^3}{6}\right) + O(x^4)$$

For $\lim_{x\to 0^+} f(x)$ to exist finitely: $\frac{k^2}{2} - m = 0 \implies m = \frac{k^2}{2}$

Under this condition: $\text{RHL} = \lim_{x\to 0^+} f(x) = \frac{k^3}{6}$

**Option (A):** For $k = 1, m = 1/2 \implies \text{RHL} = 1/6$.

Since $\text{LHL} = \text{RHL} = f(0) = d = 1/6$, $f(x)$ is continuous at $x = 0$.

**Option (B):** $\text{LHL} = \text{RHL} = 1/6 \neq d$.

Since the limit exists but does not equal $f(0)$, it is a removable isolated point discontinuity.

**Option (C):** For $k = 2, m = 2 \implies \text{RHL} = \frac{2^3}{6} = \frac{8}{6} = \frac{4}{3}$. $\text{LHL} = 1/6$.

Both limits exist finitely, but $\text{LHL} \neq \text{RHL}$. Magnitude of jump $= \left|\frac{8}{6} - \frac{1}{6}\right| = \frac{7}{6}$.

This is a non-removable jump discontinuity regardless of $d$.

**Option (D):** If $a = 0, c = 1$, and $b = 1/2$: $b - \frac{a}{2} + \frac{c}{2} = \frac{1}{2} + \frac{1}{2} = 1 \neq 0$

$N(x) = x^2 + \frac{1}{6}x^3 + O(x^4)$, so $f(x) = \frac{1}{x} + \frac{1}{6} + O(x)$ as $x \to 0^-$.

Hence, $\lim_{x\to 0^-} f(x) = -\infty$, making $x = 0$ an essential infinite discontinuity.

## Paragraph

Consider the function f : R → R defined by

$$f(x) = \begin{cases} x^m \sin\left(\frac{1}{x^n}\right) & \text{if} x > 0 \\ 0 & \text{if} x = 0 \\ -(-x)^m \sin\left(\frac{1}{(-x)^n}\right) & \text{if} x < 0 \end{cases}$$

where m ∈ R and n ∈ $R^+$.

Q9. Let n = 2. If f(x) is twice differentiable at x = 0, but f "(x) is NOT continuous at x = 0, then the complete set of values of m is:

(A) $(4, 6]$ (B) $(4, 6)$ (C) $[4, 6)$ (D) $(2, 4]$

Ans. A

Sol. Analysis of Higher Derivatives at the Origin:

For x > 0, successive differentiation yields:

$f(x) \sim x^m \sin(x^{-n})$

$f'(x) \sim x^{m-n-1} \cos(x^{-n})$ + higher power terms

$f''(x) \sim x^{m-2n-2} \sin(x^{-n})$ + higher power terms

$f^{(k)}(x) \sim x^{m-kn-k}$. (sin($x^{-n}$) or cos($x^{-n}$)) + higher power terms

General Conditions at x = 0:

$f^{(k)}(0)$ exists if and only if the lowest power in $\frac{f^{(k-1)}(x)}{x}$ is strictly positive :

$m - (k-1)n - (k-1) - 1 > 0 \quad \Rightarrow m > (k-1)n + k$

$f^{(k)}(x)$ is continuous at x = 0 if and only if the lowest power in $f^{(k)}(x)$ is strictly positive:

$m - kn - k > 0 \quad \Rightarrow m > k(n+1)$

Given n = 2:

f "(0) exists $\Leftrightarrow m > (2-1)(2) + 2 \quad \Rightarrow m > 4$.

f "(x) is continuous at x = 0 $\Leftrightarrow m > 2(2+1) \quad \Rightarrow m > 6$.

Thus, f "(0) exists but f "(x) is NOT continuous at x = 0 when $m \in (4, 6]$.

(At m = 6, f "(x) contains a term proportional to $\sin(x^{-2})$, which oscillates rapidly as $x \to 0$ without approaching a single limit, so f "(x) is discontinuous at x = 0.)

Q10. Let m = 7 and n = 1. Let g(x) = f '(x) for all $x \in R$. Which of the following statements is **TRUE**?

(A) g'(x) is continuous at x = 0, but g'(x) is not differentiable at x = 0

(B) g"(x) is continuous at x = 0, but g"(x) is not differentiable at x = 0

(C) g"(x) is not continuous at x = 0, but g"(0) exists. (D) g(x) is not differentiable at x = 0

Ans. B

Sol. Given m = 7, n = 1, and g(x) = f '(x) :

g(x) = f '(x)

g'(x) = f "(x)

g"(x) = f "'(x)

For k = 3 (which corresponds to g"(x) = f "'(x)):

The lowest power exponent in f "'(x) is m – 3n – 3 = 7 – 3 – 3 = 1.

Since 1 > 0, $\lim_{x\to 0} g''(x) = 0 = g''(0)$, *so* $g''(x)$ is continuous at x = 0.

For differentiability of g"(x) at x = 0 (g "'(0) = $f^{(4)}(0)$):

$f^{(4)}(0)$ exists $\Leftrightarrow m > (4-1)(1) + 4 = 7$.

Since m = 7, m > 7 is FALSE.

Explicitly, $g'''(0) = \lim_{x\to 0} \dfrac{g''(x) - 0}{x} = \lim_{x\to 0} \dfrac{x \cdot C \cdot \cos(x^{-1})}{x} = \lim_{x\to 0} C\cos(1/x)$, which does not exist.

Hence, g"(x) is continuous at x = 0, but g"(x) is not differentiable at x = 0.

Q1. For $n \in \mathbb{N}$, let $S_n = \sum_{k=1}^{n} \tan^{-1}\left(\dfrac{2k}{k^4 + k^2 + 2}\right)$ and let $S_\infty = \lim_{n\to\infty} S_n$. Evaluate the value of the limit: $L = \lim_{n\to\infty}\left(\dfrac{4n^3 + 7n^2 + 2}{n+3} \cdot (S_\infty - S_n)\right)$

Ans. 4

Sol. $T_k = \tan^{-1}\left(\dfrac{2k}{k^4 + k^2 + 2}\right) = \tan^{-1}\left(\dfrac{(k^2+k+1) - (k^2-k+1)}{1 + (k^2+k+1)(k^2-k+1)}\right)$

$= \tan^{-1}(k^2+k+1) - \tan^{-1}(k^2-k+1)$

$S_n = \tan^{-1}(n^2+n+1) - \tan^{-1}(1) = \tan^{-1}(n^2+n+1) - \dfrac{\pi}{4}$

$S_\infty = \lim_{n\to\infty} S_n = \dfrac{\pi}{2} - \dfrac{\pi}{4} = \dfrac{\pi}{4}$

Thus, the difference $S_\infty - S_n$ is:

$S_\infty - S_n = \cot^{-1}(n^2+n+1) = \tan^{-1}\left(\dfrac{1}{n^2+n+1}\right)$

Now, substituting $S_\infty - S_n$ into the required limit:

$$L = \lim_{n\to\infty}\left(\frac{4n^3+7n^2+2}{n+3}\cdot\tan^{-1}\left(\frac{1}{n^2+n+1}\right)\right)$$

$$= \lim_{n\to\infty}\left(\frac{4n^3+7n^2+2}{(n+3)(n^2+n+1)}\cdot\frac{\tan^{-1}\left(\frac{1}{n^2+n+1}\right)}{\frac{1}{n^2+n+1}}\right) = 4$$

Q2. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x) = ||x-1|-2|$ and let $g:\mathbb{R}\to\mathbb{R}$ be defined by $g(x) = 1 - f(2x-3)$

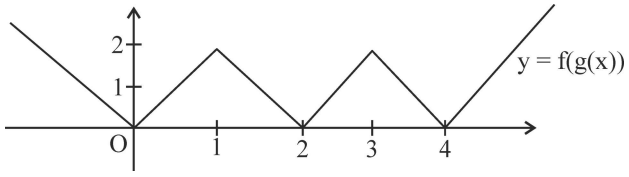
Consider the composite function $h:\mathbb{R}\to\mathbb{R}$ given by $h(x) = (f\circ g)(x)$.

If $N$ represents the total number of points in $\mathbb{R}$ where $h(x)$ is not differentiable, and $I = \int_0^4 h(x)\,dx$, then find the value of $N + I$.

Ans. 9

Sol. $g(x) = 1 - ||2x-4|-2|$

$\Rightarrow f(g(x)) = |||2x-4|-2|-2|$




N = 5

$$I = \int_0^4 h(x)dx = \left(\frac{1}{2}\times 2\times 2\right)\times 2 = 4$$

Therefore, $N + I = 5 + 4 = 9$

Q3. Let $U = \{1,2,3\}$ be the universal set. Let $S$ be the set of all ordered triples $(A,B,C)$ of subsets of $U$ that simultaneously satisfy the following three conditions:

(i) $[(A\cup B^c)\cap(B\cup C^c)\cap(C\cup A^c)]^c = U$

(ii) $(A\cup B)^c\cup(B\cup C)^c\cup(C\cup A)^c = \emptyset$

(iii) $A\cap B\neq\emptyset$, $B\cap C\neq\emptyset$, and $C\cap A\neq\emptyset$

where $X^c$ denotes the complement of set $X$ with respect to $U$. Find the number of elements in $S$.

Ans. 6

Sol. Let the 8 disjoint basic regions of the Venn diagram for subsets $A, B, C \subseteq U$ be denoted as:

$R_1 = A\cap B\cap C$; $R_2 = A\cap B\cap C^c$; $R_3 = A\cap B^c\cap C$; $R_4 = A\cap B^c\cap C^c$

$R_5 = A^c\cap B\cap C$; $R_6 = A^c\cap B\cap C^c$; $R_7 = A^c\cap B^c\cap C$; $R_8 = A^c\cap B^c\cap C^c$

**Analysis of Condition (i):** Let $X = (A\cup B^c)\cap(B\cup C^c)\cap(C\cup A^c)$. $X^c = U \implies X = \emptyset$.
Evaluating $X$, an element $x\in X$ if and only if $x\in(A\cap B\cap C)$ or $x\in(A^c\cap B^c\cap C^c)$.
Thus, $X = R_1\cup R_8 = \emptyset \implies R_1 = \emptyset$ and $R_8 = \emptyset$.

**Analysis of Condition (ii):**

$(A \cup B)^c \cup (B \cup C)^c \cup (C \cup A)^c = \emptyset$

$\Rightarrow (A \cup B)^c = \emptyset, (B \cup C)^c = \emptyset, (C \cup A)^c = \emptyset.$

This implies $A \cup B = U, B \cup C = U, C \cup A = U.$

$(A \cup B)^c = \emptyset \implies R_7 = \emptyset, R_8 = \emptyset.$; $(B \cup C)^c = \emptyset \implies R_4 = \emptyset, R_8 = \emptyset.$;

$(C \cup A)^c = \emptyset \implies R_6 = \emptyset, R_8 = \emptyset.$

Combining conditions (i) and (ii), we have:$R_1 = R_4 = R_6 = R_7 = R_8 = \emptyset$

The only remaining non-empty regions are $R_2, R_3$, and $R_5$.

**Analysis of Condition (iii):** Since $R_1 = \emptyset$:$A \cap B = R_2 \neq \emptyset$; $C \cap A = R_3 \neq \emptyset$;$B \cap C = R_5 \neq \emptyset$

Since $U = \{1, 2, 3\}$ contains 3 elements and $R_2, R_3, R_5$ are pairwise disjoint non-empty sets whose union is $U$, each region must contain exactly 1 element.

The number of ways to partition the 3 distinct elements of $U$ into the 3 labeled non-empty regions $R_2, R_3, R_5$ is: $3! = 6$

Q4. Let $A = \{1, 2, 3\}$. Find the number of relations on $A$ which are both reflexive and symmetric, but not transitive.

Ans. 3

Sol. Let $A = \{1, 2, 3\}$.

Total number of relations on $A$ that are both reflexive and symmetric is:

$2^3 = 8$

An equivalence relation is a relation that is reflexive, symmetric, and transitive.

The number of equivalence relations on $A = \{1, 2, 3\}$ equals the number of partitions of $A$ (the Bell number $B_3$):

Total equivalence relations on set A = $1 + 3 + 1 = 5$.

The number of relations that are reflexive and symmetric but NOT transitive is:

$8 - 5 = 3$

Q1. Let $A = \{1, 2, 3\}$. Let $R$ be a reflexive relation on $A$ satisfying $R \circ R^{-1} = R$, where $R^{-1}$ denotes the inverse of relation $R$ and $\circ$ denotes the composition of relations. The total number of such relations $R$ on $A$ is

Note : Let R be a relation from set A to set B ($R \subseteq A \times B$), and S be a relation from set B to set C ($S \subseteq B \times C$). The **composition of R and S**, denoted by $S \circ R$, is the relation from A to C defined as:

$S \circ R = \{(a, c) \in A \times C \mid \exists b \in B$ such that $(a, b) \in R$ and $(b, c) \in S\}$ :

Ans. 5

Sol. Given that $R$ is a reflexive relation on $A = \{1, 2, 3\}$, we have $(x, x) \in R$ for all $x \in A$. We are given $R \circ R^{-1} = R$.

**Symmetry:**

Let $(x, y) \in R$. Since $R$ is reflexive, $(x, x) \in R$.

By definition of relation composition, $(a, b) \in R \circ R^{-1}$ if there exists $c \in A$ such that $(c, a) \in R$ and $(c, b) \in R$.

Choosing $c = x$, $a = y$, and $b = x$, we have $(x, y) \in R$ and $(x, x) \in R$.

This implies $(y, x) \in R \circ R^{-1}$.

Since $R \circ R^{-1} = R$, it follows that $(y, x) \in R$.

Thus, $R$ is symmetric, which implies $R^{-1} = R$.

**Transitivity:**

Since $R^{-1} = R$, the given condition becomes $R \circ R = R$.

If $(x, y) \in R$ and $(y, z) \in R$, then by definition of composition, $(x, z) \in R \circ R = R$.

Thus, $R$ is transitive.

Therefore, $R$ is reflexive, symmetric, and transitive, which means $R$ is an equivalence relation on $A$.

The number of equivalence relations on a set with 3 elements is equal to the number of partitions of $A = \{1, 2, 3\} = 5$

Q2. If $x > 0$ is a real solution to the equation $\sum_{k=1}^{2} \tan^{-1}\left(\frac{2k}{k^4 + k^2 + 2}\right) + \sin^{-1}\left(\frac{1}{\sqrt{1 + x^2}}\right) = \frac{\pi}{4}$, then the value of $x$ is equal to

Ans. 7

Sol. Equation becomes $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{11}\right) + \sin^{-1}\left(\frac{1}{\sqrt{1 + x^2}}\right) = \frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left(\frac{3}{4}\right) + \sin^{-1}\left(\frac{1}{\sqrt{1 + x^2}}\right) = \frac{\pi}{4}$

$\Rightarrow \sin^{-1}\left(\frac{1}{\sqrt{1 + x^2}}\right) = \tan^{-1}\left(\frac{1}{7}\right) = \sin^{-1}\left(\frac{1}{\sqrt{50}}\right)$

x = 7

Q3. Let $S = \{-3, -2, -1, 0, 1, 2, 3\}$. For each $a \in S$, define

$$f(a) = \lim_{x \to 0} \left(\frac{\cos(2ax)}{\cos(a^2 x)}\right)^{\frac{1}{x^2}}$$

An equivalence relation $R$ is defined on $S$ by $aRb \iff f(a) = f(b)$. Let $k$ be the total number of distinct equivalence classes of $S$ under $R$, and let $E$ be the equivalence class containing the element $a = 2$. If $|E|$ denotes the number of elements in $E$, then find the value of $k \cdot |E| + \sum_{x \in E} x^2$.

Ans. 17

Sol. $f(a) = \exp\left(\lim_{x \to 0} \frac{1}{x^2}\left(\frac{\cos(2ax)}{\cos(a^2 x)} - 1\right)\right) = \exp\left(\lim_{x \to 0} \frac{\cos(2ax) - \cos(a^2 x)}{x^2 \cos(a^2 x)}\right)$

$$\lim_{x\to 0}\frac{\cos(2ax)-\cos(a^2x)}{x^2\cos(a^2x)}=\frac{a^4-4a^2}{2}$$

Thus, $f(a)=e^{\frac{a^4-4a^2}{2}}$

Now, evaluate $P(a)=a^4-4a^2$ for elements $a\in S$:

For $a=0$: $P(0)=0 \implies f(0)=e^0=1$

For $a=1,-1$: $P(1)=P(-1)=1-4=-3 \implies f(1)=f(-1)=e^{-3/2}$

For $a=2,-2$: $P(2)=P(-2)=16-16=0 \implies f(2)=f(-2)=e^0=1$

For $a=3,-3$: $P(3)=P(-3)=81-36=45 \implies f(3)=f(-3)=e^{45/2}$

The equivalence relation $aRb \iff f(a)=f(b)$ partitions $S$ into distinct equivalence classes:

$E_1=\{-2,0,2\};E_2=\{-1,1\};E_3=\{-3,3\}$

Hence, the total number of distinct equivalence classes is $k=3$.
The equivalence class containing $a=2$ is $E=E_1=\{-2,0,2\}$.

The number of elements in $E$ is $|E|=3$.
The sum of squares of elements in $E$ is: $\sum_{x\in E}x^2=(-2)^2+0^2+2^2=4+0+4=8$

$k\cdot|E|+\sum_{x\in E}x^2=3\cdot 3+8=9+8=17$